

BAYESIAN SHRINKAGE ESTIMATION OF RELIABILITY FROM
CENSORED SAMPLE WITH EXPONENTIAL FAILURE MODEL

Department of Statistics, Faculty of Science,
Banaras Hindu University, Varanasi 221005

Key words:

Bayes estimators; Bayesian shrinkage estimators;
exponential distribution; maximum likelihood
estimators; reliability function; shrinkage
estimators; type II censoring.

Summary:

In this paper the problem of estimation of reliability function $R(t)$ and the scale parameter θ from a type II censored sample with exponential failure distribution is considered. The problem of shrinking an unbiased estimator $\hat{\theta}$ of the parameter θ was considered by several authors (cf. Thompson (1968), Lemmer (1981), etc.). The ordinary shrunk estimator, $K\hat{\theta} + (1-K)\theta_0$, with K a constant ($0 \leq K \leq 1$), was found to be better than $\hat{\theta}$ if θ is close to the natural origin θ_0 . Here we postulate a prior distribution for the parameter around the natural origin and use the ordinary Bayes estimator in place of $\hat{\theta}$ in the ordinary shrunk estimator. Bayesian shrinkage estimators of θ and $R(t)$ have thus been obtained and Monte Carlo study of their efficiencies with respect to unbiased and maximum likelihood estimators (mle's) was undertaken and it was found that the proposed estimators are better than unbiased estimators and mle's.

1. INTRODUCTION

The simplest and most widely exploited model in the life testing problems is the one parameter exponential distribution with cumulative distribution function (cdf) given by

$$(1) \quad F_x(x/\theta) = 1 - \exp\left[-\left(\frac{x}{\theta}\right)\right]$$

$$x \geq 0, \theta > 0$$

where θ is the scale parameter and gives the average or mean life of the item under study. For a situation where the failure rate appears to be more or less constant or in case of waiting time problems the exponential distribution would be an adequate choice.

Also the reliability function, $R(t)$, which is the probability of survival for at least time t is given by

$$(2) \quad R(t) = 1 - F_x(t) = \exp\left[-\left(\frac{t}{\theta}\right)\right]$$

The mle $\hat{\theta}$ of θ and hence of $R(t)$ can easily be obtained without any complicated mathematical aid. The mle of θ is also the uniformly minimum variance unbiased estimator (UMVUE). The problem of estimation of reliability was considered by several authors (cf. Pugh (1963) and Basu (1964)). Bayes estimator of θ as well as of $R(t)$ has also been considered by several authors.

However, Thompson (1968) suggested the problem of shrinking an unbiased estimator $\tilde{\theta}$ of the parameter θ towards a natural origin θ_0 by the shrunken estimator $K\tilde{\theta} + (1-K)\theta_0$, ($0 \leq K \leq 1$), which is more efficient than $\tilde{\theta}$ if the true value of θ is close to θ_0 and less efficient otherwise. Singh and Bhatkulikar (1978) applied the significance test of validity of the prior value θ_0 and suggested to use the above shrunken estimator if the prior value is accepted and to use the unbiased estimator $\tilde{\theta}$ otherwise. The problem of estimation of the parameters through the ordinary shrinkage estimators was also considered by Mehta and Srinivasan (1971) and by Pandey (1979).

Lemmer (1981) gave the concept of using an ordinary Bayes estimator $\hat{\theta}_0$ instead of θ_0 in the ordinary shrinkage estimator $K\tilde{\theta} + (1-K)\theta_0$ and thus derived the concept of Bayesian shrinkage estimators. He considered the estimation of Binomial, Poisson and Normal parameters through Bayesian shrinkage techniques. But he has paid no attention to the life testing models. Pandey and Singh (1983) derived the Bayesian shrinkage estimator of the scale parameter θ in complete sample of exponential distribution and did not see the effect of censoring in the data. Pandey (1983) and Pandey and Upadhyay (1984) considered the same for Weibull parameters in different cases of complete sample. Shrinkage or Bayesian shrinkage techniques in reliability estimation of life testing distributions has not yet been used.

In this paper we consider the problem of estimation of the scale parameter θ and the reliability function $R(t)$ in the above exponential distribution when the information regarding θ and hence $R(t)$ is available in the form of prior distribution of θ around its prior or guess value, θ_0 . The general case has been discussed when the prior distribution of θ places a weight $(1-b)$ in the prior or guess value and distributes the rest of probability mass $b(0 \leq b \leq 1)$ over some suitable interval.

In the next section we propose the corresponding Bayesian shrinkage estimators of θ and $R(t)$ after deriving the expressions of their ordinary Bayes estimators from type II censored sample of life testing data from exponential distribution.

In the last section we obtain the relative efficiencies of these Bayesian shrinkage estimators with respect to the mle's and unbiased estimators.

2. BAYESIAN SHRINKAGE ESTIMATORS OF θ AND $R(t)$

Let us consider that n items with failure time distribution given by (1) are subjected to life testing and that the experimentation is terminated as soon as the r -th item fails ($r \leq n$). Also let $x_1 \dots x_r$ be the

observed failure times for the first r components. Then the likelihood function for these r failure times is given by

$$(3) \quad L(\underline{x}; \theta) = \frac{n!}{(n-r)!} \frac{1}{\theta^r} \exp \left[-\left(\frac{S_r}{\theta}\right) \right]$$

where

$$S_r = \sum_{i=1}^r x_i + (n-r)x_r.$$

The prior of θ , $g(\theta)$, places a weight $(1-b)$ in the prior or guess value θ_0 of the parameter θ and distributes the rest of probability mass b over some suitable interval of θ according to

$$f(\theta) = \frac{1}{\theta^c} \exp \left[-\left(\frac{a}{\theta}\right) \right] \quad a, c \geq 0, \quad \theta > 0$$

cf. Sinha & Kale (1980).

Hence

$$(4) \quad g(\theta) = \begin{cases} (b/\theta^c) \exp \left[-\left(\frac{a}{\theta}\right) \right] & \text{if } \theta \neq \theta_0, \quad \theta > 0 \\ (1-b) & \text{if } \theta = \theta_0. \end{cases}$$

The Bayes estimator of θ which is the mean of the posterior distribution of θ is, therefore, given by

$$\begin{aligned} \hat{\theta}_0 &= \frac{\int_0^\infty \theta L(\underline{x}; \theta) g(\theta) d\theta}{\int_0^\infty L(\underline{x}; \theta) g(\theta) d\theta} \\ &= \frac{\left\{ \int_0^\infty \frac{b}{\theta^c} \exp \left[-\left(\frac{a}{\theta}\right) \right] \frac{n!}{(n-r)!} \frac{1}{\theta^r} \exp \left[-\left(\frac{S_r}{\theta}\right) \right] d\theta + (1-b) \theta_0 \frac{n!}{(n-r)!} \frac{1}{\theta_0^r} \exp \left[-\left(\frac{S_r}{\theta_0}\right) \right] \right\}}{\left\{ \int_0^\infty \frac{b}{\theta^c} \exp \left[-\left(\frac{a}{\theta}\right) \right] \frac{n!}{(n-r)!} \frac{1}{\theta^r} \exp \left[-\left(\frac{S_r}{\theta}\right) \right] d\theta + (1-b) \frac{n!}{(n-r)!} \frac{1}{\theta_0^r} \exp \left[-\left(\frac{S_r}{\theta_0}\right) \right] \right\}} \end{aligned}$$

which on simplification reduces to

$$(5) \quad \hat{\theta}_0 = \frac{b \frac{\Gamma(r+c-2)}{(a+S_r)^{(r+c-2)}} + (1-b) \frac{1}{\theta_0^{r-1}} \exp\left[-\left(\frac{S_r}{\theta_0}\right)\right]}{b \frac{\Gamma(r+c-1)}{(a+S_r)^{(r+c-1)}} + (1-b) \frac{1}{\theta_0^r} \exp\left[-\left(\frac{S_r}{\theta_0}\right)\right]}$$

The mle of θ which is also the UMVUE of θ is given by (cf. Sinha & Kale (1980))

$$(6) \quad \hat{\theta} = \frac{S_r}{r}.$$

The inference about the reliability function ($R(t) = P[x \geq t] = \exp[-(\frac{t}{\theta})]$) can easily be drawn without explicit knowledge of the posterior of R because R is an increasing function of θ only. Hence the Bayes estimator of R is given by

$$\hat{R}_0 = \frac{\int_0^\infty R(t) L(\underline{x}; \theta) g(\theta) d\theta}{\int_0^\infty L(\underline{x}; \theta) g(\theta) d\theta}.$$

Putting the values of $R(t)$, $L(\underline{x}; \theta)$ and $g(\theta)$ in this equation, we get

$$\hat{R}_0 = \frac{S_1 + (1-b) \exp\left[-\left(\frac{t}{\theta_0}\right)\right] \frac{n!}{(n-r)!} \frac{1}{\theta_0^r} \exp\left[-\left(\frac{S_r}{\theta_0}\right)\right]}{S_2 + (1-b) \frac{n!}{(n-r)!} \frac{1}{\theta_0^r} \exp\left[-\left(\frac{S_r}{\theta_0}\right)\right]}$$

where

$$S_1 = \int_0^\infty \exp\left[-\left(\frac{t}{\theta}\right)\right] \frac{b}{\theta^c} \exp\left[-\left(\frac{a}{\theta}\right)\right] \frac{n!}{(n-r)!} \frac{1}{\theta^r} \exp\left[-\left(\frac{S_r}{\theta}\right)\right] d\theta$$

and

$$S_2 = \int_0^\infty \frac{b}{\theta^c} \exp\left[-\left(\frac{a}{\theta}\right)\right] \frac{n!}{(n-r)!} \frac{1}{\theta^r} \exp\left[-\left(\frac{S_r}{\theta}\right)\right] d\theta$$

which on solving reduces to

$$(7) \quad \hat{R}_0 = \frac{b \frac{\Gamma(r+c-1)}{(a+t+S_r)(r+c-1)} + (1-b) \frac{1}{\theta_0^r} \exp \left[-\left(\frac{t+S_r}{\theta_0}\right) \right]}{b \frac{\Gamma(r+c-1)}{(a+S_r)(r+c-1)} + (1-b) \frac{1}{\theta_0^r} \exp \left[-\left(\frac{S_r}{\theta_0}\right) \right]}$$

The mle of $R(t)$ is given by

$$(8) \quad \hat{R} = \exp \left[-\left(\frac{t}{\hat{\theta}}\right) \right]$$

where $\hat{\theta}(=\tilde{\theta})$ is the mle of θ . Also the UMVUE of $R(t)$ is (cf. Sinha & Kale (1980))

$$(9) \quad \tilde{R} = \begin{cases} \left(1 - \frac{t}{S_r}\right)^{r-1} & \text{if } t \leq S_r \\ 0 & \text{otherwise.} \end{cases}$$

Hence the proposed Bayesian shrinkage estimator of the scale parameter θ and the reliability function $R(t)$ is given by

$$T_\theta = K\hat{\theta} + (1-K)\hat{\theta}_0$$

and

$$(10) \quad T_R = K\tilde{R} + (1-K)$$

respectively, where $K(0 \leq K \leq 1)$ is a constant.

3. RELATIVE EFFICIENCY

We define

$$E_1 = \text{Ref}(T_R, \tilde{R}) = \frac{\text{Var}(\tilde{R})}{\text{MSE}(T_R)}$$

$$E_2 = \text{Ref}(T_R, \hat{R}) = \frac{\text{MSE}(\hat{R})}{\text{MSE}(T_R)}$$

$$E_3 = \text{Ref}(T_\theta, \tilde{\theta}) = \text{Ref}(T_\theta, \hat{\theta})$$

$$= \frac{\text{Var}(\tilde{\theta})}{\text{MSE}(T_\theta)}$$

where

$\text{Ref}(T_1, T_2)$ = Relative efficiency of an estimator T_1 with respect to T_2 .

MSE = Mean square error.

Var = Variance.

A Monte Carlo study of 1000 samples was conducted for different values of the ratio θ/θ_0 ($=0,25(0,25)2,00$) in order to study the efficiency of these Bayesian shrinkage estimators of θ and $R(t)$ with respect to their mle's and unbiased estimators (i.e. E_1 , E_2 and E_3 defined above). The results were studied in three parts. Firstly the effect was seen by varying the two constants b and K ($=0,25(0,25)0,75$) only together with θ/θ_0 . In this case all other factors were fixed at ($a = 2,00$, $c = 3,00$, $n = 10$, $r/n = 0,8$, $t = 5,00$ and $\theta_0 = 2,00$) (see table 1). Secondly the censoring fraction $\frac{r}{n}$ ($=0,2(0,2)0,8$) and the sample size n ($=10, 20$) were varied with $\theta/\theta_0 \cdot t$, the upper time limit in $F_x(t)$, was chosen as 5,00 for both $n=10$ and $n=20$. Other factors a , c and θ_0 were same as that in first part of the study and the two constants b and k were both taken to be 0,50 (see table 2).

Lastly we observe the efficiencies by varying the two constants a ($=1,2,4,8$) and c ($=1(1)4$) only. The other factors are fixed at $a = 2,00$, $c = 3,00$, $b = K = 0,50$, $\theta_0 = 2,00$ and $t = 5,00$ (see table 3). The extreme cases for θ/θ_0 ($=0,25$ and $2,00$) are not included in the tables because the efficiency in these two cases was generally found to be less than unity.

TABLE 1

Efficiency of Bayesian shrinkage estimators of θ and $R(t)$ with respect to their mle's and unbiased estimators when only b and K vary together with θ/θ_0 . Other values are fixed at

$a = 2,00$, $c = 3,00$, $n = 10$, $\frac{r}{n} = 0,8$, $\theta_0 = 2,00$, $t = 5,00$

b	K	θ/θ_0								
		0,50			0,75			1,00		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
0,25	0,25	0,09	0,14	0,30	1,71	1,67	1,94	10,30	8,62	8,70
	0,50	0,19	0,30	0,51	1,92	1,87	1,93	3,45	2,89	3,25
	0,75	0,53	0,82	0,84	1,51	1,48	1,47	1,69	1,42	1,66
0,50	0,25	0,12	0,20	0,39	1,80	1,76	1,90	6,75	5,65	5,35
	0,50	0,25	0,39	0,59	1,84	1,79	1,75	2,92	2,45	2,65
	0,75	0,60	0,93	0,85	1,44	1,41	1,37	1,60	1,34	1,54
0,75	0,25	0,19	0,30	0,54	1,80	1,75	1,77	4,07	3,41	3,17
	0,50	0,35	0,55	0,72	1,68	1,63	1,55	2,30	1,93	2,03
	0,75	0,69	1,07	0,90	1,34	1,31	1,26	1,46	1,22	1,39

b	K	θ/θ_0								
		1,25			1,50			1,75		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
0,25	0,25	3,27	2,63	2,78	1,56	1,25	1,41	1,00	0,83	0,97
	0,50	2,40	1,93	2,16	1,65	1,33	1,49	1,22	1,00	1,14
	0,75	1,54	1,24	1,48	1,38	1,11	1,30	1,21	1,00	1,15
0,50	0,25	2,84	2,29	2,34	1,54	1,24	1,36	1,05	0,87	0,99
	0,50	2,11	1,70	1,87	1,54	1,24	1,37	1,18	0,98	1,09
	0,75	1,45	1,17	1,38	1,30	1,05	1,22	1,16	0,96	1,09
0,75	0,25	2,39	1,92	1,92	1,53	1,23	1,31	1,13	0,93	1,03
	0,50	1,82	1,46	1,59	1,44	1,16	1,27	1,18	0,97	1,07
	0,75	1,35	1,08	1,27	1,23	0,99	1,15	1,13	0,93	1,06

TABLE 2

Efficiency of Bayesian shrinkage estimators of θ and $R(t)$ with respect to their mle's and unbiased estimators when sample size n and censoring fraction $\frac{r}{n}$ are varied together with θ/θ_0 . The other constants are fixed as

$a = 2,00$, $c = 3,00$, $b = K = 0,50$, $\theta_0 = 2,00$, $t = 5,00$

n	r/n	θ/θ_0								
		0,50			0,75			1,00		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
10	0,2	0,99	1,69	1,42	3,05	2,21	2,69	3,39	1,84	2,98
	0,4	0,50	0,86	0,92	2,45	2,17	2,19	3,18	2,30	2,82
	0,6	0,36	0,56	0,71	2,13	2,00	1,96	2,99	2,38	2,68
	0,8	0,25	0,39	0,59	1,84	1,79	1,75	2,92	2,45	2,65
20	0,2	0,55	0,89	0,95	2,44	2,19	2,19	3,09	2,23	2,71
	0,4	0,27	0,42	0,60	1,76	1,74	1,68	2,93	2,45	2,59
	0,6	0,21	0,31	0,51	1,43	1,43	1,42	2,80	2,47	2,54
	0,8	0,18	0,26	0,48	1,21	1,21	1,23	2,74	2,49	2,51

n	r/n	θ/θ_0								
		1,25			1,50			1,75		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
10	0,2	3,04	1,50	2,47	2,69	1,29	2,20	2,34	1,16	1,91
	0,4	2,68	1,78	2,31	2,08	1,39	1,75	1,72	1,18	1,48
	0,6	2,36	1,78	2,07	1,78	1,34	1,54	1,40	1,08	1,25
	0,8	2,11	1,70	1,87	1,54	1,24	1,37	1,18	0,98	1,09
20	0,2	2,54	1,73	2,18	2,10	1,40	1,76	1,72	1,17	1,46
	0,4	2,10	1,69	1,89	1,53	1,23	1,37	1,17	0,96	1,07
	0,6	1,86	1,61	1,72	1,25	1,08	1,16	0,95	0,83	0,91
	0,8	1,66	1,48	1,56	1,06	0,95	1,00	0,85	0,77	0,83

TABLE 3

Efficiency of Bayesian shrinkage estimators of θ and $R(t)$ with respect to their mle's and unbiased estimators when a and c are varied together with θ/θ_0 . The other factors being fixed at

$b = K = 0,50$, $n = 10$, $r/n = 0,8$, $\theta_0 = 2,00$, $t = 5,00$

a	c	θ/θ_0								
		0,50			0,75			1,00		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
1,00	1,00	0,28	0,44	0,64	1,15	1,13	1,20	1,39	1,17	1,21
	2,00	0,34	0,52	0,68	1,56	1,52	1,46	2,03	1,70	1,78
	3,00	0,36	0,56	0,69	1,75	1,71	1,51	2,61	2,18	2,22
	4,00	0,38	0,59	0,68	1,79	1,74	1,45	2,92	2,44	2,39
2,00	1,00	0,21	0,32	0,55	1,21	1,18	1,32	1,47	1,23	1,30
	2,00	0,24	0,37	0,58	1,64	1,60	1,66	2,24	1,88	2,04
	3,00	0,25	0,39	0,59	1,84	1,79	1,75	2,92	2,45	2,65
	4,00	0,26	0,41	0,59	1,88	1,84	1,71	3,29	2,75	2,91
4,00	1,00	0,15	0,23	0,44	1,33	1,29	1,52	1,64	1,37	1,45
	2,00	0,15	0,24	0,46	1,74	1,70	1,93	2,58	2,16	2,44
	3,00	0,16	0,24	0,46	1,90	1,85	2,07	3,36	2,81	3,27
	4,00	0,16	0,25	0,47	1,95	1,90	2,10	3,74	3,12	3,65
8,00	1,00	0,13	0,20	0,41	1,58	1,54	1,80	1,98	1,65	1,72
	2,00	0,13	0,20	0,42	1,85	1,80	2,08	3,04	2,54	2,90
	3,00	0,13	0,20	0,42	1,92	1,87	2,16	3,69	3,09	3,66
	4,00	0,13	0,20	0,42	1,94	1,89	2,17	3,92	3,29	3,92

a	c	θ/θ_0								
		1,25			1,50			1,75		
		E_1	E_2	E_3	E_1	E_2	E_3	E_1	E_2	E_3
1,00	1,00	1,25	1,00	0,98	1,13	0,91	0,86	1,06	0,87	0,81
	2,00	1,62	1,31	1,36	1,32	1,06	1,11	1,12	0,92	0,96
	3,00	2,00	1,61	1,75	1,51	1,22	1,35	1,18	0,98	1,09
	4,00	2,32	1,87	2,06	1,68	1,35	1,55	1,25	1,03	1,20
2,00	1,00	1,26	1,02	0,98	1,11	0,89	0,83	1,04	0,85	0,78
	2,00	1,68	1,36	1,42	1,32	1,07	1,10	1,11	0,91	0,95
	3,00	2,11	1,70	1,87	1,54	1,24	1,37	1,18	0,98	1,09
	4,00	2,46	1,98	2,24	1,72	1,39	1,60	1,26	1,04	1,22
4,00	1,00	1,29	1,04	0,98	1,08	0,87	0,79	0,98	0,81	0,72
	2,00	1,77	1,43	1,48	1,33	1,07	1,09	1,08	0,89	0,90
	3,00	2,25	1,81	2,01	1,57	1,26	1,39	1,18	0,97	1,08
	4,00	2,62	2,11	2,44	1,77	1,42	1,64	1,27	1,05	1,23
8,00	1,00	1,35	1,08	0,98	1,04	0,84	0,73	0,90	0,74	0,62
	2,00	1,91	1,54	1,56	1,35	1,08	1,07	1,04	0,85	0,84
	3,00	2,40	1,94	2,13	1,62	1,30	1,41	1,18	0,97	1,06
	4,00	2,74	2,21	2,56	1,81	1,46	1,68	1,29	1,06	1,24

The above tables show that the efficiency of these Bayesian shrinkage estimators of θ and $R(t)$ with respect to their unbiased estimators and mle's is, generally, greater than unity if θ/θ_0 lies between 0,50 and 1,75. The upper limit 1,75 may be increased in some cases but it always has to be in the neighbourhood of 2,00 except when r is small. For the cases when both n and r are very small the upper limit of the effective interval can be further increased and it can be raised even to 3,00 or more (table 2) as far as the efficiencies E_1 and E_3 are concerned. The efficiency E_2 , in table 1 for $b = K = 0,75$ (or large) or in table 2 for $n = 10$ and $r/n = 0,2$, shows that the lower limit 0,50 of the effective interval can also be shifted to left in these cases.

Apart from these results, there are also a few cases, especially with E_2 and E_3 when the upper limit 1,50 will have to be shifted back. For example, E_2 is 0,99 if $b = K = 0,75$, $\theta/\theta_0 = 1,50$ in table 1. Similarly in table 3 if $\theta/\theta_0 = 1,50$ the small values of c will provide E_2 and E_3 both less than unity.

If one desires to use the proposed Bayesian shrinkage estimators, the values of b , K , n and r/n must be decided. On the basis of a detailed study of tables 1, 2 and 3; table 4 provides a guideline to choose various constants that occur in T_θ and T_R for best possible results. In practice, however, θ is unknown and hence also θ/θ_0 . But it is evident from table 4 that as far as the choice of sample size and censoring fractions are concerned they should be as small as possible for almost all values of θ/θ_0 . If, however, the ratio θ/θ_0 is nearing unity from either side for large n and r/n also, the efficiency will remain greater than unity. If b and K are chosen as less than or equal to 0,25, T_θ and T_R will be good if the unknown value θ lies anywhere in a wide interval $\theta/\theta_0 > 0,50$.

So we observe that the proposed Bayesian shrinkage estimators of the scale parameter θ and the reliability function $R(t)$ in case of exponential distribution are better if θ/θ_0 lies between 0,50 and 1,75 and their efficiencies can be further improved if the values of constants are chosen as described in table 4.

TABLE 4

Table showing the choice of constants.

θ/θ_0	TO INCREASE	K	b	r/n	n	a	c
$\approx 0,50$	E_1	$\geq 0,75$	$\geq 0,75$	S	S	S	L
	E_2	$\geq 0,75$	$\geq 0,75$	S	S	S	L
	E_3	$\geq 0,75$	$\geq 0,75$	S	S	S	L
$\left. \begin{array}{l} > 0,50 \\ < 1,00 \end{array} \right\}$	E_1	$\leq 0,50$	$\leq 0,50$	S	S	L	L
	E_2	$\leq 0,50$	$\leq 0,50$	S	S	L	L
	E_3	$\leq 0,50$	$\leq 0,50$	S	S	L	L
$\approx 1,00$	E_1	$\leq 0,25$	$\leq 0,25$	S	S	L	L
	E_2	$\leq 0,25$	$\leq 0,25$	L	S	L	L
	E_3	$\leq 0,25$	$\leq 0,25$	S	S	L	L
$\left. \begin{array}{l} > 1,00 \\ \leq 1,25 \end{array} \right\}$	E_1	$\leq 0,25$	$\leq 0,25$	S	S	L	L
	E_2	$\leq 0,25$	$\leq 0,25$	S	S	L	L
	E_3	$\leq 0,25$	$\leq 0,25$	S	S	L	L
$> 1,25$	E_1	$\leq 0,50$	$\leq 0,50$	S	S	L	L
	E_2	$\leq 0,50$	$\leq 0,50$	S	S	L	L
	E_3	$\leq 0,50$	$\leq 0,50$	S	S	L	L

Note (1)L: large values, S: small values.

Note (2): Small values of 'a' imply the values near 1,00 whereas large 'a' means the values greater than or equal to 8,00. Similarly large 'c' means the values of 'c' greater than or equal to 4,00.

REFERENCES

- BASU, A.P. (1964). Estimates of reliability for some distributions useful in life testing. *Technometrics* **6**, 215-219.
- LEMMER, H.H. (1981). From ordinary to Bayesian shrinkage estimators. *South African Statist. J.* **15**, 57-72.
- MEHTA, J.S. and SRINIVASAN, R. (1971). Estimation of the mean by shrinking to a point. *J. Amer. Statist. Ass.* **66**, 86-90.
- PANDEY, B.N. (1979). On shrinkage estimation of normal population variance. *Commun. Statist.* **A8**, 359-365.
- PANDEY, B.N. and SINGH, K.N. (1983). Bayesian shrinkage estimation of the Exponential scale parameter, sent for publication.
- PANDEY, M. (1983). A Bayesian approach to shrinkage estimation of Weibull scale parameter, sent for publication.
- PANDEY, M. and UPADHYAY, S.K. (1984). Bayesian shrinkage estimators of Weibull parameters, sent for publication.
- PUGH, E.L. (1963). The best estimate of reliability with exponential case. *Operat. Res.* **11**, 57-61.
- SINGH, G. and BHATKULIKAR, S.G. (1978). Shrunk estimation in Weibull distribution. *Sankhya, Ser. B* **39**, 382-393.
- SINHA, S.K. and KALE, B.K. (1980). *Life testing and reliability estimation*. Wiley.
- THOMPSON, J.R. (1968). Some shrinkage techniques for estimating the mean. *J. Amer. Statist. Ass.* **63**, 113-123.